

A task-based guide to the current public workflow. Read the HTML article for links and full boundaries.

1. Prepare and specify

Put responses into long or wide data and state the covariance question.

Key functions: `traits()`, `gllvmTMBcontrol()`; see HTML

Start with ordinary covariance; add a relationship source only when the study design supplies one.

2. Fit

Fit every stacked-trait model through one entry point.

Key functions: `gllvmTMB()`

Use a supported response family and retain the formula that defines the estimand.

3. Check fit health

Check convergence, gradients, Hessian information, and fitted-response mismatch before interpretation.

Key functions: `check_gllvmTMB()`, `gllvmTMB_diagnose()`; see HTML

A syntactically valid model can still be weakly identified for its data.

4. Interpret covariance

Extract rotation-invariant covariance, correlation, and shared-variance summaries first.

Key functions: `extract_Sigma()`, `extract_correlations()`; see HTML

Loadings and ordination scores are orientation-dependent; interpret them after covariance summaries.

5. Choose a target-specific uncertainty route

Inspect available methods and their returned status rather than assuming a generic confidence interval.

Key functions: `profile_targets()`, `confint_inspect()`; see HTML

Interval availability and calibration depend on the estimand, family, and covariance tier.

6. Report, predict, or simulate

Make report-ready tables and figures, predict fitted responses, or simulate new responses.

Key functions: `extract_Sigma_table()`, `plot_Sigma_heatmap()`; see HTML

Name the target and scale in every reported table or figure.

Typical route: Prepare and specify -> Fit -> Check fit health -> Interpret covariance. Prediction, simulation, and uncertainty are conditional branches.

Navigation aid only: a documented function can still be unsuitable for a particular family, data set, or covariance tier.